

Artur Rodrigues Sousa Alves, 22 years old

Software Developer | Full Stack Developer | A.I

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience developing full-stack applications, architecting distributed systems, and building high-impact generative AI platforms. Graduated in Software Engineering from the University of Brasília (UnB), with specialization in applied Artificial Intelligence and scalable production systems. Expert in implementing RAG (Retrieval-Augmented Generation) systems, integrating LLMs from multiple providers (GPT-4, Claude, Gemini), and orchestrating multi-agent systems using CrewAI and LangChain. Proven track record developing robust and scalable solutions with modern stack (React, Node.js, Python, PostgreSQL) that increased productivity by 45% through intelligent automation and advanced prompt engineering. Hands-on experience with vector databases (Pgvector, FAISS), semantic embeddings, document chunking, retrieval optimization, and event-driven architectures. Strong foundation in distributed systems concepts, data structures, algorithms, and Site Reliability Engineering (SRE). Hands-on experience with AWS cloud infrastructure (ECS, Lambda, S3, RDS), containerization with Docker, CI/CD pipelines, and modern DevOps practices. Passionate about transforming complex challenges into innovative, scalable, and efficient solutions. Proven skills in leading production projects, delivering measurable results focused on business metrics, and building highly available systems that scale to thousands of simultaneous users.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Dart, C#, Java, HTML5, CSS3.

Frameworks: React, React Native, Vue.js, Angular.js, Node.js, Flutter, .NET, Redux Toolkit, Django, FastAPI.

AI & Data: RAG, Vector Databases (Pgvector, FAISS), LLM Integration (GPT-4, Claude, Gemini), Embeddings, Prompt Engineering, LangChain, CrewAI, Multi-Agent Systems, NLP, TensorFlow, PyTorch, n8n. 45% increase in productivity with AI.

Databases: PostgreSQL, MySQL, Redis, Firebase, Vector Databases.

DevOps: Git, GitHub, GitLab, Docker, CI/CD, AWS (ECS, Lambda, S3, RDS).

Methodologies: Scrum, Kanban, Code Review, Pair Programming.

Testing: E2E, Integration, Black Box/White Box, Vitest, Jest. Software Testing Teaching Assistant at UnB.

Languages: Portuguese (native), English (C1), Spanish (A2).

EXPERIENCE AND ACADEMIC PROJECTS

Thesis: RAGBot - Intelligent Search System for UnB Academic Documents

August 2025 - December 2025

University of Brasília (UnB)

Architected RAG system with Pgvector, Gemini 2.5 pro API, custom document chunking/ingestion pipeline, and semantic embeddings, achieving 85% accuracy, 70% reduction in search time. Full-stack SaaS platform with Python, Angular.js/TypeScript, PostgreSQL, tests with 80%+ coverage, scaling to 500+ paying subscribers and 3,000 active users.

Full Stack Developer – IZE Business Management

September 2025 - January 2026

Full Stack Developer building scalable interfaces and APIs with AI. Worked with TypeScript, Python, PostgreSQL, and Firebase, implementing LLM chatbot that reduced response time by 40% and increased team productivity by 30% via prompt engineering. Automated 15+ workflows with n8n + OpenAI API.

Full Stack Developer - Acesso UnB DACES/UnB Project

April 2024 - January 2025

University extension project with Flutter/Dart and Node.js/MySQL. Implemented AI for voice recognition and NLP achieving 92% accuracy, mapping 50+ routes with intelligent navigation. Mobile app to assist autonomous mobility for people with disabilities.

Full-Stack Developer - UnB/MEPA Energy

November 2024 - March 2025

Development with React, TypeScript, Docker, Python, and Django. Implemented ML models for predictive analysis of energy data, improving forecasting accuracy by 25% through feature engineering.

Front-End Developer - Pagglo, Brasília (Internship)

January 2024 – May 2025

iOS/Android development with React Native, TypeScript, and JavaScript. Integration with C#/.NET APIs. Implemented fraud detection system with ML reducing false positives by 35% and recommendation engine that increased user engagement by 28%. Agile methodologies (Scrum).

Full-Stack Developer - UnB/DNIT

March 2023 - July 2023

Developed web solutions using Front-end and back-end technologies: .NET, C#, React, TypeScript, PostgreSQL, Git, GitHub, Figma.

EDUCATION

Software Engineering

July 2021 – December 2025

UnB, University of Brasília

CERTIFICATIONS AND COMPLEMENTARY COURSES

Multi AI Agent Systems with crewAI Course Completion Certificate – Jan 2025 – Feb 2025

NLW Agents Advanced Event Participation Certificate by RocketSeat – July 2025

RocketSeat NLW AI Course Attendance Certificate – Sept 2023

NLW Expert React Native Event Participation Certificate by RocketSeat – Feb 2024